

# # VPC Gateway Endpoint

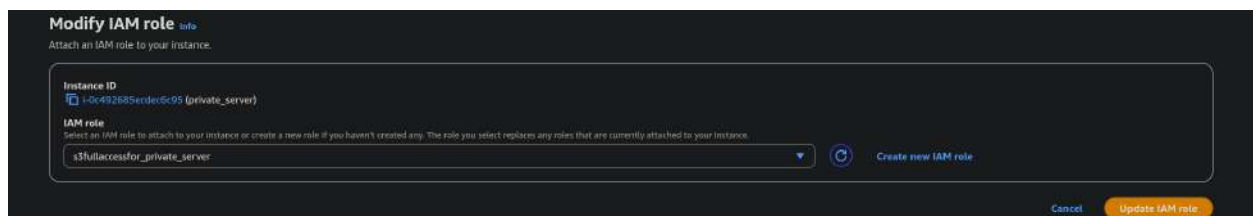
## #objective:

The objective of this lab is to understand how a **Gateway VPC Endpoint** works and how it allows private communication between resources inside a VPC and supported AWS services (Amazon S3 and DynamoDB) without using an Internet Gateway, NAT Gateway, or public internet.

## #Lab Environment

- 1 Amazon VPC
- 1 Private Subnet
- 1 Public Subnet
- 1 Amazon EC2 Instance
- Amazon S3 Bucket
- Route Tables
- Gateway VPC Endpoint

# First I launch 2 instances: one in private subnet and another in public subnet. Then I SSH into the private instance and try to list the S3 bucket, but I can't even attach the `s3FullAccess` role to this private instance.



```
/u/
[ec2-user@ip-172-31-48-194 ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2081ms

[ec2-user@ip-172-31-48-194 ~]$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-48-194 ~]$ aws s3 ls
```

# Create VPC Endpoint Gateway for give securely access s3 service , in creation mode there select VPC, route table and service

The first screenshot shows the 'Route tables (1/2)' page. It displays a table with columns: Name, Route Table ID, Main, and Associated Id. The 'private-route' is selected, with Route Table ID 'rtb-077744c3997905c8a' and associated with 'subnet-0f27584a154c4b81 (RD5-Pvt-subnet-2)'. A note at the bottom states: 'When you use an endpoint, the source IP addresses from your instances in your affected subnets for accessing the AWS service in the same region will be private IP addresses, not public IP addresses. Existing connections from your affected subnets to the AWS service that use public IP addresses may be dropped. Ensure that you don't have critical tasks running when you create or modify an endpoint.'

The second screenshot shows the 'Services (1/8)' page. It displays a table with columns: Service Name, Owner, and Type. The 'com.amazonaws.ap-south-1.s3' service is selected, which is of type 'Gateway'.

The third screenshot shows the 'Endpoints (1/1)' page. It displays a table with columns: Name, VPC endpoint ID, Endpoint type, Status, Service name, and Service network. The 's3endpoint' is shown with VPC endpoint ID 'vpce-0f42fb2c9cd2740fd', type 'Gateway', status 'Available', service name 'com.amazonaws.ap-south-1.s3', and service network '-'. Below this, the 'Route tables (1)' section shows the 'private-route' selected, with Route Table ID 'rtb-077744c3997905c8a' and associated with 'subnet-0f27584a154c4b81 (RD5-Pvt-subnet-2)'. A 'Manage route tables' button is visible.

# After these configuration i try to again list s3 bucket then finally i can list then s3 bucket privately without going through internet

```
aws s3 ls
2026-07-02 14:30:35 angela-s3-bucket-1
2026-06-28 20:55:27 pk-it
2026-06-21 17:36:27 saa.bishnuthapa.info.np
```

#Conclusion:

In this lab, I successfully created an **AWS Gateway VPC Endpoint** for Amazon S3. I verified that an EC2 instance located in a private subnet could securely access an S3 bucket without requiring an Internet Gateway or NAT Gateway. This exercise demonstrated how

Gateway VPC Endpoints provide secure, private, and cost-effective access to supported AWS services while ensuring that traffic remains within the AWS network.